

1

Quantification

1.1 Define the Measurand (Analyte)

The initial question for the analyst is to define what is expected to be measured. According to the International Vocabulary of Metrology [1], the “quantity intended to be measured”¹ is called the measurand, or more specifically, the analyte, when considering measurement methods applied to chemical and biochemical substances. But this simple definition may be misleading while an analyte may have variable forms during the analytical process. It is not always certain that the substance finally measured is initially intended to be measured. For example, during sample preparation, the initial organic form of the analyte may change to inorganic, and what was intended to be measured is finally modified. For instance, in living organisms, heavy metal is present combined with proteins, such as mercury to metallothionein. Still, when analyzed after mineralization, it can be transformed into sulfate, perchlorate, or nitrate.

A well-known catastrophic example is the Minamata disease; when looking for mercury in food samples, the oldest methods were based on the complete sample mineralization to obtain mercury nitrate. Soon after, it was realized that the toxic forms of mercury were organic derivatives. Hence, so-called total mercury had no great toxicological interest compared to the different organic forms. Speciation techniques in mineral analysis or chiral chromatographic methods are good examples of innovative approaches devoted to better maintaining the analyte in its expected form. Therefore, quantification in analytical sciences is often less straightforward than claimed. From the metrological point of view, the difficult traceability of chemical substances to international standards is one of these obstacles.

This is detailed in Section 6.3 as an introduction to the estimation of measurement uncertainty (MU) among many other sources of uncertainty. The encapsulated conception of modern and highly computerized instruments may also prevent the analyst from assessing what is measured. Digits displayed on the instrument screen represent what is “intended to be measured.” The paradoxical consequence is that discussing the true nature of the analyte is often avoided, while more attention

1 Definitions or quotations extracted from standards or official documents are between double quotes.

should be paid to this question. The goal of this chapter is to propose things to consider on this topic. Many examples are based on mass spectrometry (MS) hyphenated methods because several are now considered highly compliant from a metrological point of view.

1.1.1 Quantification and Calibration

The metrology motto could be measuring is comparing. Therefore, when quantifying an analyte, the comparison principle must be previously defined. This preliminary step is usually called calibration. In modern analytical sciences, most methods use measuring instruments ranging from simple, specific electrodes to sophisticated devices; therefore, calibration procedure may enormously vary according to the nature of the instrumentation. This chapter attempts to classify the different quantification/calibration strategies applied in analytical laboratories. Because this subject is not harmonized, the employed vocabulary may vary from one domain of analysis to another and be confusing. For each term, we tried to give a definition, but it may be incomplete due to the considerable number of analytical techniques. Many suggested definitions are listed in the glossary at the end of the book.

Whatever the measuring domain, classic differences are made between direct and indirect measurement techniques. Direct method can usually refer to a measurement standard, for instance, when measuring the weight of an object on a two-pan balance with standard weights. Indirect measurements are performed using a transducer, a “device, used in measurement, which provides an output quantity with a specified relation to the input quantity.”

Reversely, with a one-pan balance, measurements are indirect. At the same time, result is obtained by means of a mathematical model linking the calibrated piezoelectrical effect on the beam to the weight. In analytical sciences, methods are usually indirect. Some exceptions are set apart, classified as direct primary operating procedures by BIPM (Section 4.2.1). For most chemical or biological analytical techniques, the measuring instrument must be calibrated with known reference items before use. Finally, quantification involves three elements, as outlined in Figure 1.1:

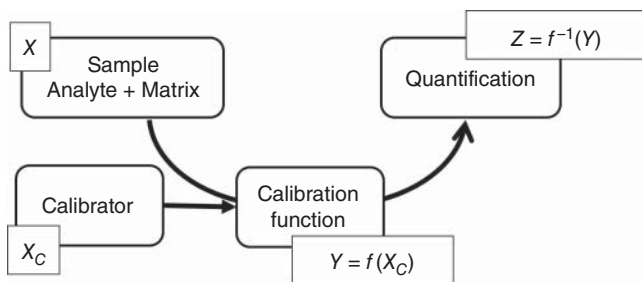


Figure 1.1 Schematic representation of the quantification principle.

- The analyte is in the working sample. Its concentration is denoted X . The searched compound (chemical or biological) is embedded within the sample matrix. It is only before any treatment that the analyte is present in the intended form. The role of sample preparation is to eliminate a large part of the matrix and concentrate on the analyte. But it may change the analyte chemical form; for instance, with the speciation of organic forms of heavy metals, sample preparation is quite different from classic mineralization.
- The calibration items are also called *calibration standards* or *calibrators*. They are prepared by the analyst to contain a known amount of a calibrant as similar as possible to the analyte. To underline this difference, it is denoted X_c . The selection of the adequate calibrant is a key-issue of quantification extensively addressed in the rest of this chapter.
- The calibration function that links the instrumental response Y to the known quantity X_c , denoted $Y = f(X_c)$.

Figure 1.1 is an attempt to recapitulate a generic quantification procedure. Most of the time, calibrators are artificially prepared and used to build the calibration function f which generally is *inverted* when analyzing an unknown sample. The three elements may be subjected to variations. Mathematical notation underlines the dissimilar roles they play for the statistical modeling of calibration and possible relationships that link the instrumental signal to the calibrant concentration. Denoting Z the predicted concentration of a sample emphasizes the role of inverting calibration function as discussed in Section 2.1. Finally, considering a given calibration dataset, distinct functions f can be fitted. A principal issue will be to select the best one because it deeply affects the global method performance. The goal of the present chapter is to describe some classical or new quantification procedures.

1.1.2 Authentic versus Surrogate

To be explicit, it is convenient to define some terms. If the chemical substance sought in the sample is called *authentic*, obviously, for many methods it is possible to prepare the calibrators with the authentic analyte. But other quantification methods exist based on a different calibration compound, which will be called surrogate standard or calibrant. It would be paradoxical to call it surrogate analyte, whereas the analyte can only be authentic. Therefore, when the analyte and the calibrant are different, it is necessary for the analyst to cautiously verify if they have equivalent analytical behavior and define an eventual adjustment method, such as a correction factor.

The measuring instrument is a transducer that converts the amount or the concentration of a chemical substance into a signal – usually electrical – according to a physical or chemical principle. How quantitative analyses are achieved varies from simple color tests for detecting anions and cations through complex and expensive instrumentation for determination of trace amounts of a compound or substance in a complex matrix. Increasingly, such instrumentation is a hybrid of techniques for separation and detection that requires extensive data processing.

The subject of analytical sciences has become so wide that complete coverage, providing clear information to an interested scientist, can only be achieved in a multi-volume encyclopedia. For instance, Elsevier published in 2022 the volume n°98 of the *Comprehensive Analytical Chemistry* handbook started in the 1980s.

The major obstacle in analytical sciences is the structural or chemical differences that exist between the analyte present in the working sample and the substance used as a calibrant. The instrument signal may depend on the authentic or surrogate structure of the analyzed substance: this dependence is marked with modern instrumentation such as mass spectrometers. On the other hand, the analyte present in a working sample is embedded with other chemicals, customarily called a matrix by the analysts. It is not always possible or easy to use the sample matrix when preparing the calibrators. These remarks lead to the definitions of four different quantification elements that can be combined to prepare or selecting calibrators and consequently obtain the calibration curve:

Authentic analyte	The same molecule or substance present in the working sample may be available for calibrator preparation, considering a high degree of purity.
Surrogate standard or calibrant	This is a reference substance that is assessed and used as a reasonable substitute for the authentic analyte. For instance, in bioanalysis, it is frequent to have metabolites or derivatives of the analyte that must be quantified without the reference molecule. Labeled molecules used in many methods involving isotopic dilution have recently been considered appropriate calibrants.
Authentic matrix	The simplest situation for using an authentic matrix is to prepare calibrants by spiking test portions of the working sample. For some applications, such as drug control, it is also possible to prepare synthetic calibrants with the same ingredients as the products to be controlled.
Surrogate matrix	This medium is considered and used as a substitute for the sample matrix. For instance, bovine serum is used in place of human serum. Then, it is assumed its behavior should be similar to the authentic matrix throughout the analytical process, including sample preparation and instrumental response.

When the surrogate matrix does not behave as the authentic or when calibration is achieved without the sample matrix, matrix effects may produce bias of trueness, as explained in Section 4.1.3. More precisely, calibration standards can be prepared with several classes of matrices. Matrix classification is widely based on analyst expertise and depending on the application domain, matrix grouping is extremely variable. For instance, broad definitions applicable to biological analysis can be as follows: